

Read [Joy Buolamwini and Timnit Gebru, 2018. "Gender shades: Intersectional accuracy disparities in commercial gender classification." Conference on fairness, accountability and transparency](<http://proceedings.mlr.press/v81/buolamwini18a/buolamwini18a.pdf>), then use it to help answer the following questions.

a. Buolamwini and Gebru use PPV and FPR as metrics to measure fairness. Find the definition of these in the paper, then look up the corresponding definition for NPV and FNR (these appear in the slides). Assuming binary labels and predictions, describe each of these four metrics as a probability, e.g. "The PPV gives the probability that the label is **BLANK**** given that the model predicts ****BLANK****."**

In the "Gender shades" paper by Buolamwini and Gebru, PPV (Positive Predictive Value) and FPR (False Positive Rate) are used as metrics to measure fairness. PPV is defined as the probability that a true label is positive given that the model predicts positive, while FPR is defined as the probability that a true label is negative given that the model predicts positive. In addition, the paper's slides also mention NPV (Negative Predictive Value) and FNR (False Negative Rate) as related metrics. NPV is the probability that a true label is negative given that the model predicts negative, while FNR is the probability that a true label is positive given that the model predicts negative. These metrics are all defined for binary labels and predictions.

b. Assuming you were applying for a loan and you know a ML classifier is deciding whether to grant it to you: would you rather have that decision made by a system with a high PPV or a high NPV? Why?

I would prefer the decision to be made by a system with a high PPV (Positive Predictive Value) rather than a high NPV (Negative Predictive Value).

A high PPV means that the probability of being granted a loan given that the model predicts I will receive one is high. In other words, the model is more likely to correctly identify those who will be granted a loan, rather than falsely identifying those who will not be granted a loan as qualified. On the other hand, a high NPV means that the probability of not being granted a loan given that the model predicts I will not receive one is high. This is important, but less relevant to me as an individual seeking a loan. It would be more problematic if the model falsely identified me as unqualified and denied me a loan, even though I was actually qualified.

c. What recommendations do Buolamwini and Gebru make regarding accountability and transparency of ML systems? How does this relate to specific metrics such as PPV or FPR?

One key recommendation is to evaluate these systems for fairness and to report on fairness metrics, such as PPV (Positive Predictive Value), FPR (False Positive Rate), and other related measures, for different subgroups of the population. Specifically, they recommend reporting

these metrics separately for different gender and skin tone subgroups, as their research showed that commercial facial recognition systems exhibited significant accuracy disparities across these subgroups.

Buolamwini and Gebru also recommend that machine learning systems be audited for bias and that transparency be increased by making the source code and training data publicly available. This would enable researchers to identify and correct potential biases in the system and to evaluate its overall fairness and accuracy.

d. What is *intersectional* about the analysis conducted by the authors? What does that analysis show?

The analysis conducted by Buolamwini and Gebru in their paper is intersectional because it examines how different subgroups of people are affected by biases and inaccuracies in machine learning systems, based on the intersection of multiple identity categories such as gender and skin tone.

Specifically, their analysis shows that commercial facial recognition systems exhibited significant accuracy disparities across gender and skin tone subgroups. They found that these systems were more likely to misclassify darker-skinned individuals, and that the accuracy disparities were most pronounced for darker-skinned females. This demonstrates the importance of considering intersectional identities when evaluating the fairness and accuracy of machine learning systems, as biases and inaccuracies can disproportionately affect certain subgroups of people based on their multiple intersecting identities.

The analysis highlights the need for greater attention to intersectionality in machine learning research, and the need for more diverse and inclusive data sets to ensure that these systems are accurate and fair for all people, regardless of their gender, race, or other intersecting identities.

e. In Section 4.7, the authors say that their "findings ... do not appear to be confounded by the quality of sensor readings." What do they mean by "confounded" in this context? Why is it important to the thesis of this paper to check whether their findings are confounded?

In the context of the authors, "confounded" refers to a situation where an observed association between two variables (in this case, the accuracy of facial recognition and skin tone or gender) is actually due to a third variable that is not being measured or controlled for.

For example, it could be argued that the accuracy disparities observed in facial recognition systems are not due to the systems themselves, but rather to the fact that darker-skinned individuals are more likely to be photographed under poor lighting conditions, which could impact the quality of the sensor readings. However, the authors state that they have controlled for this possibility by using a diverse set of images that were captured under a variety of lighting conditions, and that their findings cannot be explained by variations in the quality of sensor readings.

It is important to check whether their findings are confounded by other variables because if they are, it would weaken the conclusions that can be drawn from the analysis. If the observed

accuracy disparities were actually due to factors such as lighting or sensor quality, it would suggest that the problem is not with the facial recognition systems themselves, but rather with the conditions under which the images were captured. However, by ruling out potential confounding variables, the authors are able to more confidently attribute the observed accuracy disparities to biases in the facial recognition algorithms themselves. This strengthens the thesis of the paper and reinforces the need for greater attention to fairness and accuracy in machine learning systems.